

Spring Core & Maven

Exercise 1: Configuring a Basic Spring Application

pom.xml :

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.30</version>
  </dependency>
</dependencies>
```

BookRepository.java :

```
package com.library.repository;

public class BookRepository {
    public void display() {
        System.out.println("Book Repository Created");
    }
}
```

BookService.java :

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showService() {
        System.out.println("Book Service Created");
        bookRepository.display();
    }
}
```

applicationContext.xml :

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>


```

```

        <bean id="bookService"
            class="com.library.service.BookService">
            <property name="bookRepository"
                ref="bookRepository"/>
        </bean>
    </beans>

```

MainApp.java :

```

package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class MainApp {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "applicationContext.xml");

        BookService service =
            (BookService) context.getBean("bookService");

        service.showService();
    }
}

```

Output :

```

Book Service Created
Book Repository Created

```

Exercise 2: Implementing Dependency Injection

BookRepository.java :

```

package com.library.repository;

public class BookRepository {
    public void display() {
        System.out.println("Book Repository Connected");
    }
}

```

BookService.java :

```

package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    // Setter Injection

```

```

        public void setBookRepository(BookRepository bookRepository) {
            this.bookRepository = bookRepository;
        }

        public void showService() {
            System.out.println("Book Service Running");
            bookRepository.display();
        }
    }
}

```

applicationContext.xml :

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

    <bean id="bookService"
          class="com.library.service.BookService">
        <property name="bookRepository"
                  ref="bookRepository"/>
    </bean>
</beans>

```

LibraryManagementApplication.java :

```

package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "applicationContext.xml");

        BookService service =
            (BookService) context.getBean("bookService");

        service.showService();
    }
}

```

Output :

```

Book Service Running
Book Repository Connected

```

Exercise 4: Creating and Configuring a Maven Project

pom.xml :

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>
  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0</version>

  <dependencies>
    <!-- Spring Context -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.30</version>
    </dependency>

    <!-- Spring AOP -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-aop</artifactId>
      <version>5.3.30</version>
    </dependency>

    <!-- Spring Web MVC -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-webmvc</artifactId>
      <version>5.3.30</version>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.8.1</version>
        <configuration>
          <source>1.8</source>
          <target>1.8</target>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

Project Structure :

LibraryManagement

```
|  
|-- src/main/java  
|  
|-- src/main/resources  
|  
■■■ pom.xml
```

Maven Commands USED :

```
mvn clean  
mvn compile  
mvn package
```

Output :

```
BUILD SUCCESS
```
